

# AFRILANG Stdlib

## std/c/sigfft

**Français.** Bibliothèque AFRILANG 'std/c/sigfft' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : dft, magnitude, powerSpectrum, dominantFreq, inverseDft, bandEnergy.

```
'import "std/c/sigfft"' · 'use sigfft'
```

```
- 'dft(signal list number) → list number' - 'magnitude(signal list number) → list number' - 'powerSpectrum(signal list number) → list number' - 'dominantFreq(signal list number) → number' - 'inverseDft(spectrum list number, n number) → list number' - 'bandEnergy(signal list number, lo number, hi number) → number'
```

**English.** AFRILANG library 'std/c/sigfft' (tier complex, category complex). Exports 6 function(s): dft, magnitude, powerSpectrum, dominantFreq, inverseDft, bandEnergy.

```
'import "std/c/sigfft"' · 'use sigfft'
```

```
- 'dft(signal list number) → list number' - 'magnitude(signal list number) → list number' - 'powerSpectrum(signal list number) → list number' - 'dominantFreq(signal list number) → number' - 'inverseDft(spectrum list number, n number) → list number' - 'bandEnergy(signal list number, lo number, hi number) → number'
```

|                       |                      |               |
|-----------------------|----------------------|---------------|
| Catégorie / Category  | Modules complex (c/) |               |
| Niveau / Tier         | complex              | June 16, 2026 |
| Fonctions / Functions | 6                    |               |

## Contents

## Français

### 1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/sigfft' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : dft, magnitude, powerSpectrum, dominantFreq, inverseDft, bandEnergy.

```
'import "std/c/sigfft"' · 'use sigfft'
```

```
- `dft(signal list number) → list number`
- `magnitude(signal list number) → list number`
- `powerSpectrum(signal list number) → list number`
- `dominantFreq(signal list number) → number`
- `inverseDft(spectrum list number, n number) → list number`
- `bandEnergy(signal list number, lo number, hi number) → number`
```

### 2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/sigfft"
use sigfft
```

Niveau : complex · Catégorie : Modules complex (c/)  
Domaines avancés – graphes, NLP, réseaux complexes.

### 3. Référence API

- dft(signal list number) → list•  
magnitude(signal list number) → list•  
powerSpectrum(signal list number) → list•  
dominantFreq(signal list number) → number•  
inverseDft(spectrum list number, n number) → list•  
bandEnergy(signal list number, lo number, hi number) → number

### 4. Exemples d'utilisation

```
import "std/c/sigfft"
use sigfft
```

```
create result = dft(/* args */)
say result
```

```
create result = magnitude(/* args */)
say result
```

```
create result = powerSpectrum(/* args */)
say result
```

```
create result = dominantFreq(/* args */)
say result
```

```
create result = inverseDft(/* args */)
say result
```

```
create result = bandEnergy(/* args */)
say result
```

## 5. Bonnes pratiques

- Préférez ``import "std/c/sigfft"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

## 6. Annexe — source

module sigfft

```
function dft(signal list number) returns list number  
end
```

```
function magnitude(signal list number) returns list number  
end
```

```
function powerSpectrum(signal list number) returns list number  
end
```

```
function dominantFreq(signal list number) returns number  
end
```

```
function inverseDft(spectrum list number, n number) returns list number  
end
```

```
function bandEnergy(signal list number, lo number, hi number) returns number  
end  
end
```

## English

### 1. Overview

AFRILANG library 'std/c/sigfft' (tier complex, category complex). Exports 6 function(s): dft, magnitude, powerSpectrum, dominantFreq, inverseDft, bandEnergy.

'import "std/c/sigfft"' · 'use sigfft'

```
- `dft(signal list number) → list number`
- `magnitude(signal list number) → list number`
- `powerSpectrum(signal list number) → list number`
- `dominantFreq(signal list number) → number`
- `inverseDft(spectrum list number, n number) → list number`
- `bandEnergy(signal list number, lo number, hi number) → number`
```

### 2. Installation & import

Add the module to your program:

```
import "std/c/sigfft"
use sigfft
```

Tier: complex · Category: Complex modules (c/)  
Advanced domains – graphs, NLP, complex networks.

### 3. API reference

- dft(signal list number) → list•  
magnitude(signal list number) → list•  
powerSpectrum(signal list number) → list•  
dominantFreq(signal list number) → number•  
inverseDft(spectrum list number, n number) → list•  
bandEnergy(signal list number, lo number, hi number) → number

### 4. Usage examples

```
import "std/c/sigfft"
use sigfft
```

```
create result = dft(/* args */)
say result
```

```
create result = magnitude(/* args */)
say result
```

```
create result = powerSpectrum(/* args */)
say result
```

```
create result = dominantFreq(/* args */)
say result
```

```
create result = inverseDft(/* args */)
say result
```

```
create result = bandEnergy(/* args */)
say result
```

## 5. Best practices

- Prefer ``import "std/c/sigfft"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

## 6. Appendix — source

module sigfft

```
function dft(signal list number) returns list number
end
```

```
function magnitude(signal list number) returns list number
end
```

```
function powerSpectrum(signal list number) returns list number
end
```

```
function dominantFreq(signal list number) returns number
end
```

```
function inverseDft(spectrum list number, n number) returns list number
end
```

```
function bandEnergy(signal list number, lo number, hi number) returns number
end
end
```

## Référence rapide / Quick reference

- Import : `import "std/c/sigfft"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/sigfft/>

## Source (extrait)

```
module sigfft

function dft(signal list number) returns list number
end

function magnitude(signal list number) returns list number
end

function powerSpectrum(signal list number) returns list number
end

function dominantFreq(signal list number) returns number
end

function inverseDft(spectrum list number, n number) returns list number
```

```
end
```

```
function bandEnergy(signal list number, lo number, hi number) returns number
```

```
end
```

```
end
```